

Nmap Network Scanner Report

What does it do?

It is a project that shows:

- a) Which devices are on our network?
- b) Which ports are open?
- c) What services are running?
- d) Are there risky or outdated services exposed?
- e) What should be fixed?

It is a network scanning and vulnerability assessment report using Nmap, identifying exposed services, documenting risks and recommending remediation steps.

Building

Phase 1 - Understanding the Environment.

First, we need to understand basic terms that will be helpful in our building journey.

IP vs Ports:

- Consider an IP as an address of our device in a network. Ports are in fact the apartments that are connected to our address. Two different things, yet very similar.

Why is an open port a high risk?

- An open port could be considered a high risk to our network because if the service running on that port is outdated, misconfigured or weak attackers could gain unauthorized access very easily.

Why making a project with Nmap is more valuable than just saying “I know how Nmap works”?

- Having a complete project using Nmap proves the understanding of fundamental cybersecurity concepts such as networks, risk assessments, network protocols, etc.

Phase 2 - Setting up an Ubuntu Server

To begin testing and creating this project, I installed VirtualBox and set up an Ubuntu Server VM to have as a target machine. Afterwards, with the commands **ip addr** and **ss -tuln** I managed to get informed about the VM's IP address and the TCP/UDP services that were listening at its network.

Furthermore, having confirmed the fact that the target machine and the scanning machine were on the same network, pinging was done to prove that our destination is reachable.

I even installed OpenSSH to the ubuntu machine, so that a service would actively run and use a port, for my Nmap scan to be successful.

Problems during tests:

First, I was put up against the fact that my VM was in a NAT network. This means that all network protocols and settings are managed by VirtualBox and that the VM is connected to a network in VirtualBox, and VirtualBox is connected to my device's network. The virtual machine was reconfigured from NAT to a Bridget Adapter so that it would obtain an IP address from the local network and behave as a separate host.

Later, I encountered a problem with my first scan. After running **nmap <ip>**, even though our destination was reachable and with ssh running, it returned that 0 hosts were up. I tried **nmap -Pn <ip>** which skips host discovery and proceeds with the scan, but it also did not work.

What worked was the **nmap -sT -Pn -p 22 <ip>** command in which:

- **-sT** performs a TCP Connect Scan by completing the three-way-handshake to determine whether a port is open.
- **-Pn** says to skip host discovery
- **-p** specifies the port we are looking for. In this situation, ssh uses port 22.

Phase 3 – Testing

I already talked about how I connected and verified which service was running on port 22. But it still is part of the process. Here is a full list of tests run on my Ubuntu (target) machine from my PC.

Note: I do not use the -p option here, because the only running service is SSH on port 22. If we had multiple services running on open ports, we would have to use -p.

- **nmap -sT -Pn <ip>**, which shows us the TCP connections on our destination host.
- **nmap -sT -sV -Pn <ip>**, added the -sV option, which informs us of the version of the running services.
- **nmap -sT -Pn --script "ssh-*" <ip>**, this command executes all available NSE SSH scripts. We performed service enumeration to gather additional information about the SSH service. The collected information was then used to assess whether any known vulnerabilities or security weaknesses were present.

By enumerating we mean:

- Authentication methods.
- Password brute-force testing (Only if permitted).
- Finding if public keys are accepted.
- Finding available host-keys.
- Evaluating vulnerability risks.

Phase 4 – Findings

During this procedure I found:

Running service: OpenSSH

Version: OpenSSH 10.2p1

Port: 22/TCP

Risk: Low

Evidence Port 22 responded to TCP Connect Scan. Version detection identified OpenSSH 10.2p1. No obvious vulnerabilities or weak credentials were identified during the performed enumeration.

Recommendations: Continue patch management. Use strong passwords. Prefer key-based authentication. Disable root login if not required. Restrict SSH access from using firewall rules.